A Simple Liquid Can Stop Cavities -- No Drilling Needed

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A cavity does not have to mean drilling. A simple liquid, painted onto a tooth in seconds, can stop tooth decay in young children and a major clinical trial has now confirmed it works.

The liquid is called silver diamine fluoride, or SDF. Instead of drilling out the damaged part of a tooth and placing a filling, a dentist paints SDF onto the cavity with a tiny sponge-tipped **applicator**. There is no drilling, no injection, and no **sedation**.

Dentists call this **arresting** tooth decay: the damaged area stops getting worse. SDF does not replace tooth material that is already lost. It works through two ingredients: silver helps fight the bacteria linked to decay, while fluoride helps strengthen the tooth's **mineral** structure and resist further damage.

The Phase III trial, published in JAMA Pediatrics, included 830 children younger than 6. The children were recruited through dental clinics, pediatric medical practices, and Head Start and Early Head Start programs in Michigan, New York, and Iowa. Margherita Fontana, professor of dentistry at the University of Michigan School of Dentistry and the study's lead investigator, said the treatment is very effective and safe, even in children as young as 1.

Tooth decay is the most common chronic disease among children, affecting more than 40% of American children. Left alone, it can bring severe toothaches, infections, trouble eating and sleeping, and missed school days. For many small children, regular dental treatment is hard: some cannot sit still, and others feel real fear.

There is one tradeoff. SDF permanently turns the decayed part of a treated tooth dark, which can show most on front teeth. The change is mainly **cosmetic**. For many families, skipping pain and drilling may matter more.

SDF is not new. Dentists have used it for decades in many countries, and in the United States the FDA approved it in 2014 as a medical device for tooth sensitivity. Dentists have also used it **off-label** to stop cavities. What was missing was large-scale evidence from U.S. children a key step toward FDA approval for that use.

Word Treasure

arresting -- Stopping something from getting worse.

sedation -- Medicine that makes a person sleepy or calm during treatment.

applicator -- A small tool used to put something onto a surface.

mineral -- A hard natural material that helps make up a tooth.

cosmetic -- About how something looks, rather than how it works.

off-label -- Using an approved medicine or device for a different use than the one it was first approved for.

Background you need

A cavity is a damaged spot on a tooth that gets worse when germs keep attacking it.

Dentists usually fix cavities by drilling out the damaged part and filling the hole, which can be scary or painful for small children.

Silver diamine fluoride offers a different approach: it is painted on to stop the damage from spreading, even though it cannot rebuild tooth that is already gone.

Because it needs no drilling or sleep medicine, it can help children who find normal dental visits hard.

Break it down

LEAD: A paint-on liquid can stop tooth decay in young children without drilling
KEY EVENT: A Phase III trial has confirmed SDF works
+- WHAT: SDF is painted onto a cavity with a tiny sponge-tipped applicator
+- WHO: 830 children younger than 6 in Michigan, New York, and Iowa
+- HOW: Silver fights decay-linked bacteria while fluoride strengthens the tooth
+- VOICE: Lead investigator Margherita Fontana calls it very effective and safe
EVIDENCE: Tooth decay affects more than 40% of American children and can bring toothaches, infections, and missed school days
TRADEOFF: SDF permanently darkens the decayed part, mainly a cosmetic change
CONTEXT: SDF has been used for decades elsewhere and was FDA-approved in 2014 for tooth sensitivity
OPEN QUESTION: Whether the FDA will approve SDF for stopping cavities

Why it matters

Tooth decay is the most common chronic disease among children, so a painless, drill-free treatment could spare many young children toothaches, infections, and missed school days.

Test what you remember

1. What does SDF do to tooth decay?

- (A) It drills out the damaged part of the tooth
- (B) It stops the damaged area from getting worse
- (C) It replaces tooth material that is already lost
- (D) It whitens the whole tooth

2. How is SDF put on a tooth?

- (A) It is mixed into drinking water
- (B) It is painted on with a tiny sponge-tipped applicator
- (C) It is given as a shot in the arm
- (D) It is swallowed as a pill

3. How many children took part in the Phase III trial, and how old were they?

- (A) 830 children younger than 12
- (B) More than 40% of American children
- (C) 830 children younger than 6
- (D) 300 children younger than 6

4. Who led the study?

- (A) Margherita Fontana
- (B) The U.S. Food and Drug Administration
- (C) The Head Start program
- (D) JAMA Pediatrics

5. What is one tradeoff of using SDF?

- (A) It only works on front teeth
- (B) It permanently turns the decayed part of the tooth dark
- (C) It requires sedation
- (D) It causes severe toothaches

6. What was missing before this trial?

- (A) FDA approval for tooth sensitivity
- (B) Large-scale evidence from U.S. children
- (C) Use of SDF in other countries
- (D) A way to paint SDF on teeth

Answer key (try the quiz first!): 1: B 2: B 3: C 4: A 5: B 6: B

Questions to think about

1. Margherita Fontana, the study's lead investigator: Says the treatment is very effective and safe, even in children as young as 1.

2. Many families: May care more about skipping pain and drilling than about the darkening, which is mainly cosmetic.

MY THOUGHTS (at least 20 words):